MIL-A-22305C(SHIPS)
16 October 1963
SUPERSEDING
MIL-A-22305B(SHIPS)
14 February 1962
(See Section 6.3)

MILITARY SPECIFICATION

AMPLIFIER, DISTRIBUTION AM-2123/U

1. SCOPE

1.1 This specification covers requirements of a transistorized radio frequency (r.f.) amplifier used in shipboard standard-frequency distribution systems. Amplifier channels are provided for 5 megacycles (mc.), 1 mc. and 100 kilocycles (kc.) signals. Power gain and isolation between channels are provided.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

J-C-175 - Cable Assembly, Power, Electrical, (3-Wire, 3-Prong, Grounding Plug Connector, for 125-Volt Equipment).

MILITARY

MIL-L-3661 - Lampholders; Lights, Indicator; Indicator-Light Housings; and Lenses, Indicator Light, General Specification For.
MIL-P-15024 - Plates, Identification—Information and Marking for Identification of Electrical, Electronic and Mechanical Equipment.
MIL-E-16400 - Electronic Equipment, Naval Ship and Shore: General Specification.
MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts; Preparation for Delivery of.
MIL-R-22732 - Reliability Requirements for Shipboard and Ground Electronic Equipment.

STANDARDS

MILITARY

MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-189 - Racks, Electrical Equipment, 19-Inch and Associated Panels.
MIL-STD-242 - Electronic Equipment Parts (Selected Standards).
MIL-STD-736 - Unitized Equipment Design.

DRAWINGS

BUREAU OF SHIPS

REA49028 - UG-625B/U Connector Receptacle, Series BNC.

PUBLICATIONS

BUREAU OF SHIPS

NAVSHIPS 93820 - Handbook for the Prediction of Shipboard and Shore Electronic Equipment Reliability.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 5820

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:51Z
Check the source to verify that this is the current version before use.

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N. Y.).

3. REQUIREMENTS

3.1 Qualification.- The equipment furnished under this specification shall be a product which has been tested, and passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable qualified products list.

3.2 General requirements.- The equipment shall be in accordance with all provisions of MIL-E-16400, except as otherwise specified herein.

3.2.1 Reliability.- The following is an outline of the general reliability requirements specified in MIL-E-16400 as it is applicable to this equipment.

3.2.1.1 Level.- The reliability goal for this equipment shall be 10,000 hours mean-time-between-failure (MTBF).

3.2.1.2 Reliability evaluation shall be performed in accordance with the procedures for "Reliability Assurance by Analysis and Prediction" as specified in MIL-R-22732.

3.2.1.3 Testing.- For test purposes, demonstration of attainment of a reliability level of 300 hours MTBF with a 90 percent confidence level will be required. (See 4.1.3)

3.2.1.4 In addition to the regular reliability reports of MIL-R-22732, reliability prediction reports shall be submitted whenever the design changes or additional design data becomes available affecting the prediction.

3.3 Mechanical design.-

3.3.1 Modular construction.- The equipment shall be designed for modular construction in accordance with MIL-STD-736, except that test points and test features are not required. Twelve amplifier channels shall be provided, each channel consisting of one modular assembly with its input and output connections. Module size shall be in accordance with table I. The equipment power supply may be mounted directly to the chassis, or it may be partially or wholly contained in plug-in assemblies.

Table I - Assembly size.

Inches		
Height	Length	Width
3.0	3.0	0.75

3.3.2 Channel frequency selection.- The 12 channels to be provided shall be arranged so that any of the channels can serve any of 3 frequencies, 5 mc, 1 mc, and 100 kc. Individual assemblies shall be designed for operation at one of these frequencies. Frequency service for a given channel shall be selected by inserting an amplifier assembly of the desired frequency. No switching or wiring changes shall be required to select a channel frequency. The equipment shall be capable of use with any number of assemblies from 0 to 12 plugged in, with normal operation on the channels in use.

3.3.3 Number and frequency of channels.- The equipment shall be supplied with 4 assemblies serving each of the 3 frequencies, unless otherwise specified in the contract or order. Input characteristics shall be such that any number of assemblies from 0 to 12 may be plugged in to serve any of the 3 frequencies.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:51Z
Check the source to verify that this is the current version before use.

3.3.4 Connectors.-

3.3.4.1 Assembly connectors.- Requirements of MIL-STD-736 apply, except that power and signal input and output connections shall be made thru the same connector for each assembly. Input No. 1 shall be considered 5 mc. input, likewise No. 2 shall be 1 mc., and No. 3 shall be 100 kc. Outputs from all assemblies shall be connected thru pins identified with output No. 1.

3.3.4.2 Chassis connectors.- Signal input and output connectors type UG-625 B/U, in accordance with REA49028 shall be located at the rear of the equipment. One input connector shall be for each of the 3 frequencies. Connection shall be made to furnish input signals of the 3 frequencies to the appropriate pins on all chassis receptacles. One output connector shall be provided for each amplifier assembly, and shall be connected to one assembly only.

3.3.4.3 Connector identification.- All signal connectors on the chassis shall be marked to indicate input or output, and the frequency served. Output connectors shall be numbered from 1 thru 12. Each connector shall be conveniently identified to the assembly it is serving.

3.3.5 Enclosure and mounting.- The equipment shall be capable of 19-inch rack mounting and separate shelf mounting. The outer enclosures shall be dripproof in accordance with MIL-STD-108, and may be removable for rack mounting of the equipment. Drawer slides shall be provided for withdrawing the chassis from the enclosure or rack.

3.3.6 Size.- The total enclosed volume shall not exceed one-third cubic foot. Limiting dimensions for rack mounting shall be as specified in MIL-STD-189. The overall dimensions of the equipment with outer enclosure shall not exceed the dimensions shown in table II.

Table II - Overall dimensions.

Width	21 inches
Height	8-3/4 inches
Depth	8 inches

3.3.7 Weight.- The total equipment shall not exceed 16 pounds.

3.3.8 Accessibility.- All parts shall be readily accessible for test (with equipment in operation), and for replacement or repair without injury to or removal of other parts or wiring. Chassis withdrawal and access to all parts and assemblies shall be from the front of the equipment.

3.3.9 Temperature class.- The equipment shall meet the class 4 temperature requirements.

3.3.10 Shock class.- For shock tests, the equipment shall be classified as grade A, class I, light weight, type A.

3.4 Electrical design.-

3.4.1 Duty.- The equipment shall be capable of continuous operation.

3.4.2 Power source.- The equipment shall be designed to operate from either 115 volts, 50 to 400 cycle, single phase, or 22 to 30 volts direct current (d.c.). Connection to the 115 volt source shall be thru a cable assembly permanently attached to the rear of the equipment and conforming to Type II of J-C-175. A suitable connector shall be provided on the rear of the equipment for connection to the d.c. source. Circuitry to the d.c. power connector shall be such that an external battery will furnish standby power, with no break in operation of the equipment, in case of a 115 volt power failure. The internal power supply shall not load the external battery during such a power failure.

3.4.2.1 Power indicator.- An indicator light shall be provided on the front panel in accordance with lampholder style LH74LC13CN conforming to MIL-L-3661, and associated neon lamp, to indicate "power on".

3.4.2.2 Blown-fuse indicators.- Blown-fuse indicators shall be provided in accordance with MIL-STD-242.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:51Z
Check the source to verify that this is the current version before use.

3.4.3 Input characteristics.-

3.4.3.1 Frequency.- Three separate input channels shall be provided: one each for 5 mc., 1 mc., and 100 kc.

3.4.3.2 Level.- The equipment shall be designed for operation at any input level between 0.5 and 5.0 volts (V). A manual adjustment for different input levels will be acceptable if necessary.

3.4.3.3 Impedance.- Input impedance on each channel shall be no less than 50 ohms with 12 amplifier assemblies connected to that channel.

3.4.4 Output characteristics.-

3.4.4.1 Level.- Output level shall be between 4.0 and 5.0 V measured across any load impedance between 50 ohms and 1 megohm.

3.4.4.2 Bandwidth.- The bandwidth of each frequency shall be less than 2 percent of input frequency for 3 decibels (db) down, and 10 percent for 30 db down.

3.4.4.3 Distortion.- Any single distortion product shall be at least 80 db below signal level, except harmonics shall be at least 60 db down, on each signal line.

3.4.4.4 Isolation.- The signal at any output connector shall be unaffected by a short circuit on any number of other output circuits.

3.4.5 Metering.- A front panel meter with suitable switching shall be provided for selective monitoring of operating conditions of important circuits, including power inputs and all outputs. A means shall be provided to indicate proper meter readings for all circuits monitored, either a green area on the meter, or an information plate, type A or B in accordance with MIL-P-15024, attached to the front panel.

3.4.6 Controls.- No power switch shall be provided. The panel meter selector switch shall be continually required; the input level controls if used shall be infrequently required and have suitable locking devices.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of tests.- Qualification tests as specified herein, and quality conformance inspection as specified in MIL-E-16400 shall be performed on the equipment.

4.1.2 Test procedure approval.- Procedures for all required inspections, including descriptions of test equipment to be used, shall be submitted to the bureau or agency concerned for approval at least 30 days before tests are scheduled to begin.

4.1.3 Reliability testing.- Reliability tests shall be performed as specified in MIL-R-22732 for test level number 3. The test time on each equipment shall be not less than 450 hours and not more than 720 hours. If an accept decision cannot be made before 720 hours on an equipment, the test shall be terminated and the required reports of the tests shall be submitted to the bureau or agency concerned for the decision to accept or reject. With all inputs energized, three output lines shall be monitored, one at each frequency, for level across 50 ohms. Violating the limits of 4.0 to 5.0 volts shall constitute a failure. If input level controls are used, they may be set initially, but may not be adjusted during the test. Reliability testing shall be conducted on one out of every 25 units produced.

4.2 Qualification tests.[1/]- Qualification tests shall be conducted at a laboratory satisfactory to the Bureau of Ships. Qualification tests shall consist of the tests specified in table III and 4.4. Three complete equipments shall be submitted for qualification inspection.

[1/] Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.2 and 6.2.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:51Z
Check the source to verify that this is the current version before use.

Table III - Qualification tests.

Inspection	Reference
Group I	
Surface examination	MIL-E-16400
Operating test	MIL-E-16400
Weights and dimensions	3.3.6 and 3.3.7
Heat	MIL-E-16400
Enclosure	MIL-STD-108 and 3.3.5
Power line voltage and frequency	MIL-E-16400 and 3.4.2
Group II	
Signal level	4.4.1
Bandwidth	4.4.2
Impedance	4.4.3
Isolation	4.4.4
Distortion	4.4.5
Group III	
Temperature	MIL-E-16400
Humidity	MIL-E-16400
Shock and vibration	MIL-E-16400
Inclination	MIL-E-16400
Salt spray test	MIL-E-16400
Accelerated life	MIL-E-16400

4.3 Quality conformance inspection.- Quality conformance inspection shall consist of the following:

(a) Production inspection.
(b) Production control inspection.
(c) Environmental tests.

4.3.1 Production inspection.- Production inspection shall be made on every equipment offered for delivery and shall comprise such examination and tests as will prove the workmanship and reveal possible omissions and errors in the production process. Production inspection shall include the tests shown in table IV.

Table IV - Production inspection.

Inspection	Reference
Surface inspection	MIL-E-16400
Operating test	MIL-E-16400
Signal level	4.4.1

4.3.2 Production control inspection.- Production control inspection shall be conducted on equipments which have passed the inspection of 4.3.1. Inspection shall be conducted to detect any deterioration in the design by wear of such items as dies, molds, jigs, fixtures, and to detect deviations in the material or part processing, selection, or assembly, and to detect any deterioration in the performance of the equipment not discernible under the inspection of 4.3.1. As equipments are produced and their number increases, sample units shall be selected in accordance with table V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:51Z
Check the source to verify that this is the current version before use.

Table V - Sampling for production control inspection.

Cumulative number of equipments produced	Cumulative number of equipments to be tested after 0, 1, 2 or 3 equipments have failed in any test			
	0	1	2	3
1 to 5	2	3	4	5
6 to 10	3	5	7	9
11 to 15	5	8	11	14
16 to 25	8	12	16	21
26 to 40	10	15	20	25
41 to 65	12	18	24	30
66 to 95	14	20	28	35
96 to 145	17	25	34	42
146 to 200	20	30	40	50
201 to 290	24	35	50	55
291 to 440	30	42	60	70

4.3.2.1 Each of the units selected in 4.3.2 shall be subjected to the production control tests specified in MIL-E-16400 (except water cooling, accelerated life and weld tests) and those in Group II of table III.

4.3.2.2 The number of sample equipments which shall be inspected shall be increased for each failure, as shown in table V. For example: Assume that 40 equipments have been produced, 10 of which have been subjected to the production control inspection specified herein, and all of these equipments have passed every test. From the next 25 produced, two shall be tested, provided both of them pass every test; but if one equipment fails, six additional equipments shall be selected from the increment of 25. If two of these fail in any test, another six shall be tested bringing the total number tested to 24. Thereafter, sampling shall be continued in accordance with table V.

4.3.2.3 After any change has been made in accordance with 4.3.2.4 sampling shall be started with the first entry in table V and progress downward as for a new equipment design.

4.3.2.4 If at any time the number of equipments failing in any test exceeds the total number allowed, the contractor shall be required to make such changes as may be necessary to meet the requirements of this specification. If the remedial action required to correct the equipment performance is a change of part(s), circuit lay-out, or the mechanical equivalent, such action shall be reported immediately to the bureau or agency concerned. It shall then be determined by the bureau or agency concerned whether the correction shall be made in the equipments already delivered under the contract or order.

4.3.3 Environmental tests.- Environmental tests shall be conducted on one out of every 25 equipments produced.

4.3.3.1 Disposition of shock-tested equipment.- Shock-tested samples shall be thoroughly reconditioned by the contractor before resubmitting it as a production equipment.

4.4 Test procedures.- All tests shall be performed with 12 assemblies plugged in (4 on each frequency) except as otherwise required in 4.4.3 for that test only.

4.4.1 Signal level.- An input signal of 0.5 volt at the appropriate frequency shall be applied to each of the three inputs simultaneously. All outputs shall be terminated in a 50 ohm resistor. The voltage across each load resistor shall be between 4.0 and 5.0. If a manual adjustment is provided, all output voltages shall be within these limits for one position of the adjustment. The above shall be repeated with an input level of 5.0 volts. The preceding shall be repeated with all outputs terminated in 1 megohm.

4.4.2 Bandwidth.- An input signal of 0.5 volt at the appropriate frequency shall be applied to each input simultaneously. All outputs shall be terminated in a 50 ohm resistor, and shall be adjusted (if adjustment is provided) to between 4.0 and 5.0 volts. Input frequencies shall be varied with their levels held constant. Output voltage shall be measured and plotted for output numbers 1, 3, 5, 7, 9, 11 and shall be within limits of

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:51Z
Check the source to verify that this is the current version before use.

3.4.4.2. This test shall be repeated with input signals set at 5.0 volts, all outputs terminated in 1 megohm, and voltages measured on outputs 2, 4, 6, 8, 10, 12.

4.4.3 Impedance.- Input impedance of each channel shall be measured with 12 assemblies connected across the channel being measured. Requirements of 3.4.3.3 apply.

4.4.4 Isolation.- All inputs shall be energized simultaneously with 3.0 volt signals. Output voltage shall be measured at three output connectors (one at each frequency) simultaneously, first with remaining outputs open, then with three outputs (one at each frequency) simultaneously shorted. Change in amplitude shall not exceed 1 percent on any of the measured outputs.

4.4.5 Distortion.- This test shall be run 4 times, with input voltages (three input lines energized simultaneously) and output terminations (all outputs) as specified in table VI. For each run, all output voltages shall be measured at all parts of the frequency spectrum where distortion products are likely to occur. Particular attention shall be paid to harmonics of the signal frequencies and the power frequency. Distortion shall be within limit of 3.4.4.3.

Table VI - Output terminations.

Run	Input	Output terminate
A	0.5 volt	50 ohm
B	5.0	50
C	0.5	1 megohm
D	5.0	1 megohm

4.4.6 Humidity.- Measurement and definition of failure during the humidity test shall be the same as that specified in 4.1.3.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of shipboard repair parts.-

5.1.1 Basic equipment.-

5.1.1.1 Preservation and packaging.- Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing.- Packing shall be accomplished in a manner which will insure acceptance by common carrier and will afford protection against physical or mechanical damage during handling and direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

5.1.1.3 Marking.- Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's practice. The information shall include nomenclature, contract or order number, contractor's name and destination.

5.1.2 Shipboard repair parts.- Repair parts shall be preserved and packaged Level A, packed Level C and marked Level A or C in accordance with MIL-E-17555.

5.2 Domestic shipment and storage or overseas shipment requirements.-

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made by a Government activity (see 6.1).

5.2.1.1 Preservation, packaging, packing and marking.- The equipment, accessories, repair parts and technical publications shall be preserved and packaged Level A or C, packed Level A or B and marked level A or C in accordance with MIL-E-17555.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:51Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Number and operating frequency of plug-in assemblies required, if other than 4 each for 3 frequencies (total 12) (see 3.3.3).
(c) Preservation, packaging, packing or marking requirements other than those required by 5.1 (see 5.2).

6.2 With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List QPL-22305, whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Bureau of Ships, Department of the Navy, Washington 25, D. C., and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.2.1).

6.2.1 Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

6.3 CHANGES FROM PREVIOUS ISSUE.- THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Preparing activity:
Navy - Ships
(Project 5820-N374Sh)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:51Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT

$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4063 WHICH IS OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:51Z
Check the source to verify that this is the current version before use.